



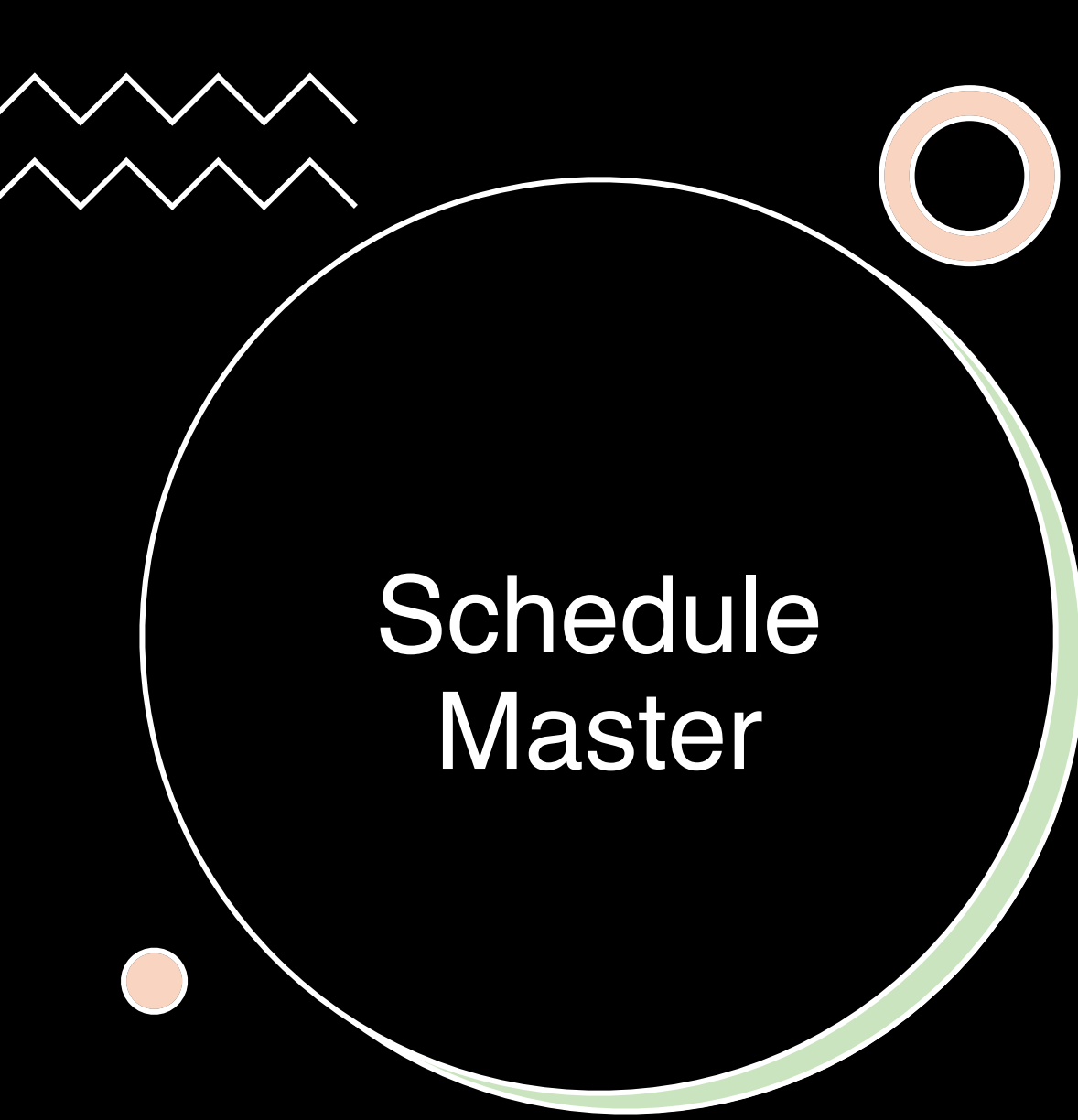
Schedule Master

Team 8 members:



Kien Mai, Arshdeep Singh, Quang
Anh Le, Jack Nguyen

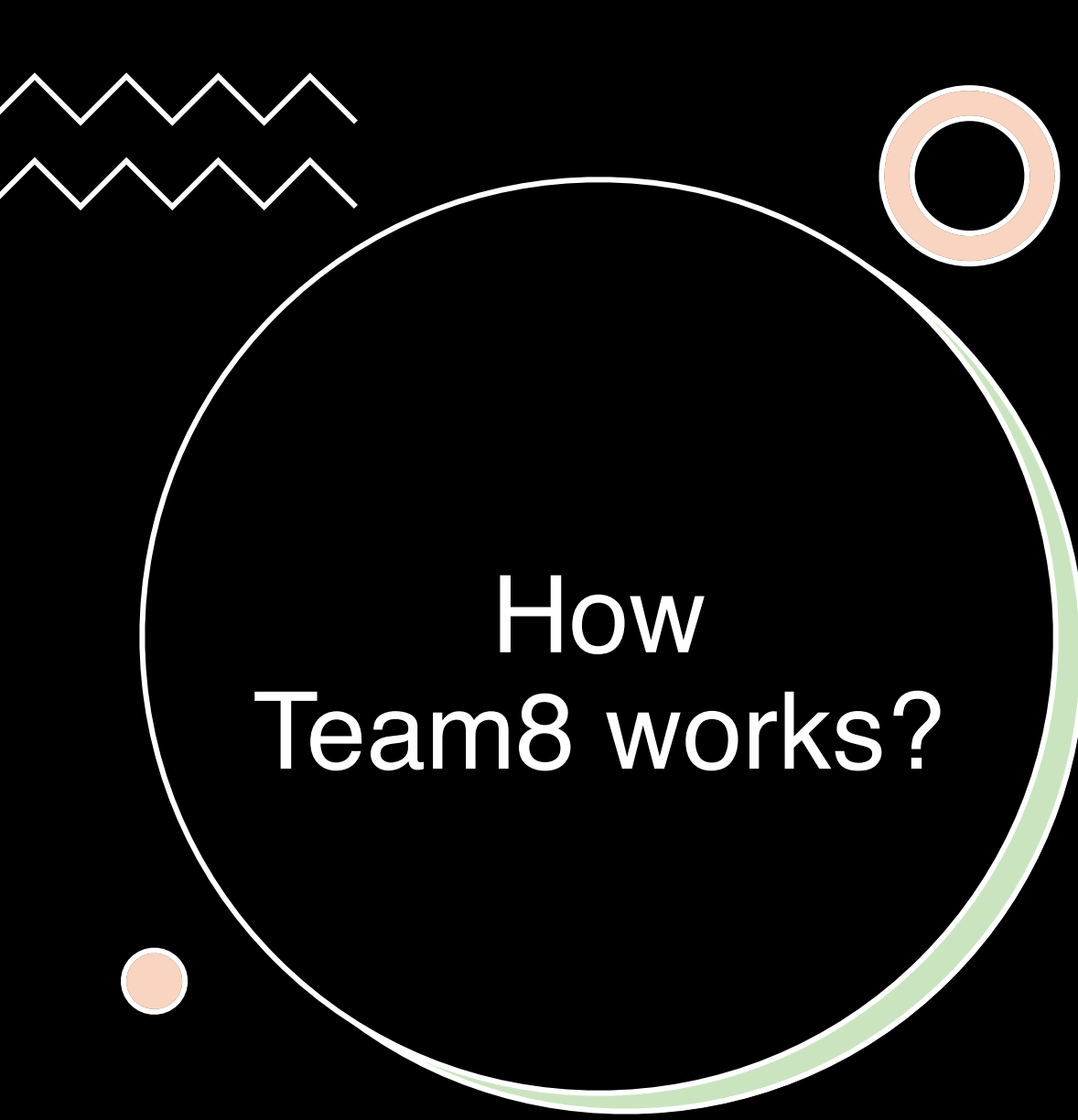




Schedule Master

- A centralized hub for all student needs
- Streamlined course management
- Seamless schedule viewing
- Efficient academic planning





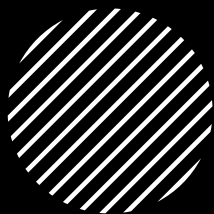
How Team8 works?

- We're all full-stack developer, frontend designer, tester, code reviewer
- Each responsible for a core feature & has full control of its implementation





Kien



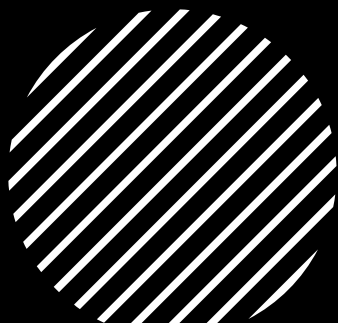
- Docker + CI/CD setup
- Roadmap feature





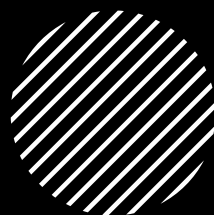
- Authentication
- Account management

Quang Anh





Jack

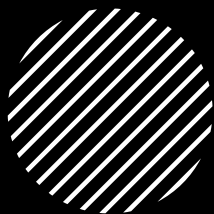


- Main page
- Database creation
- Course Management



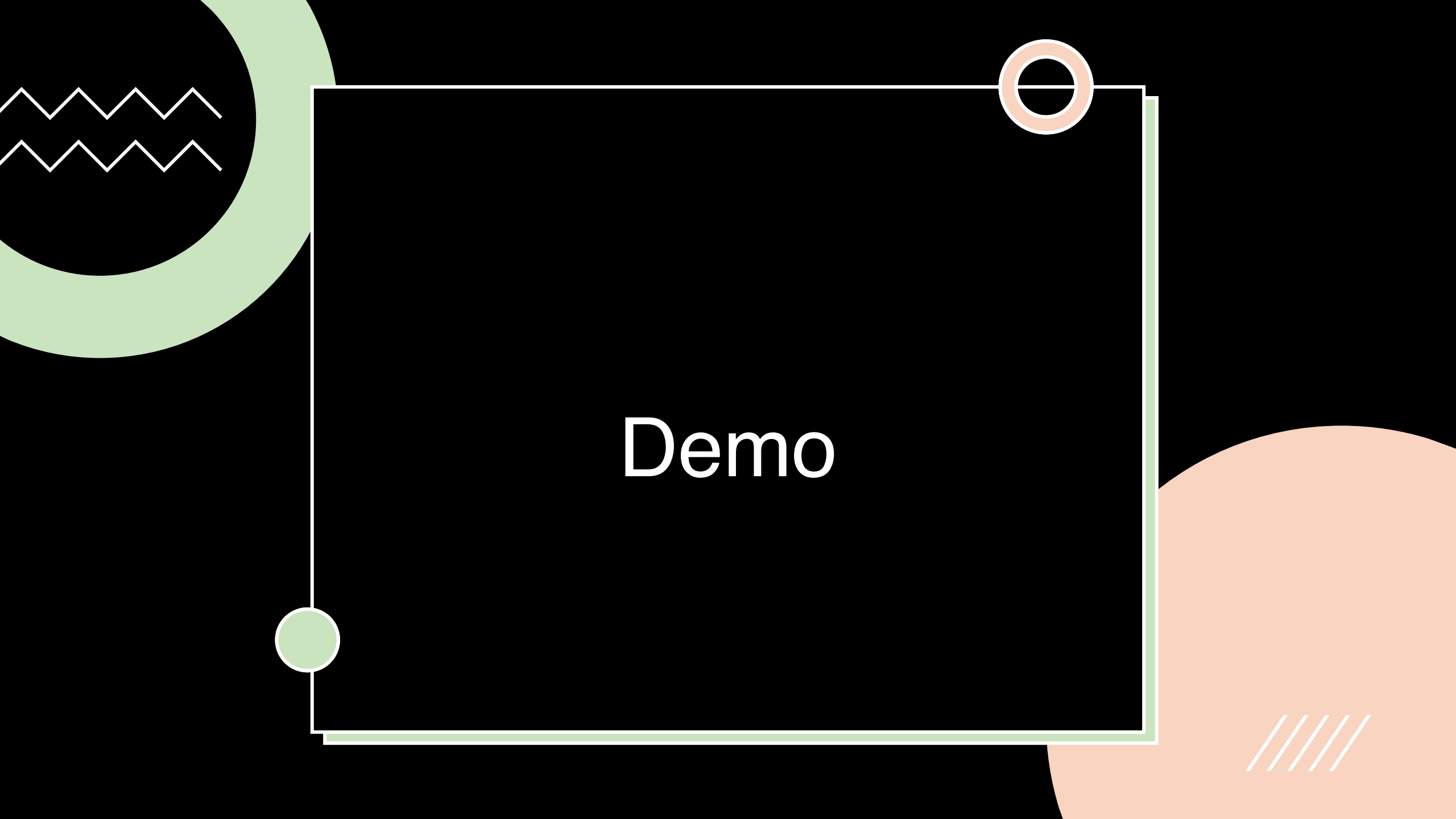


Arshdeep



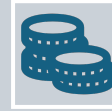
- Calendar feature
- Frontend testing using the React testing library



The image features a dark background with various geometric elements. In the top-left corner, there is a light green circle containing two white zigzag lines. A large, thin white rectangle is positioned in the center, with a light green circle at its bottom-left corner and a light orange circle at its top-right corner. The word "Demo" is written in white, sans-serif font in the center of this rectangle. In the bottom-right corner, there is a light orange shape with several white diagonal lines.

Demo

Development process



Feature Development Strategy



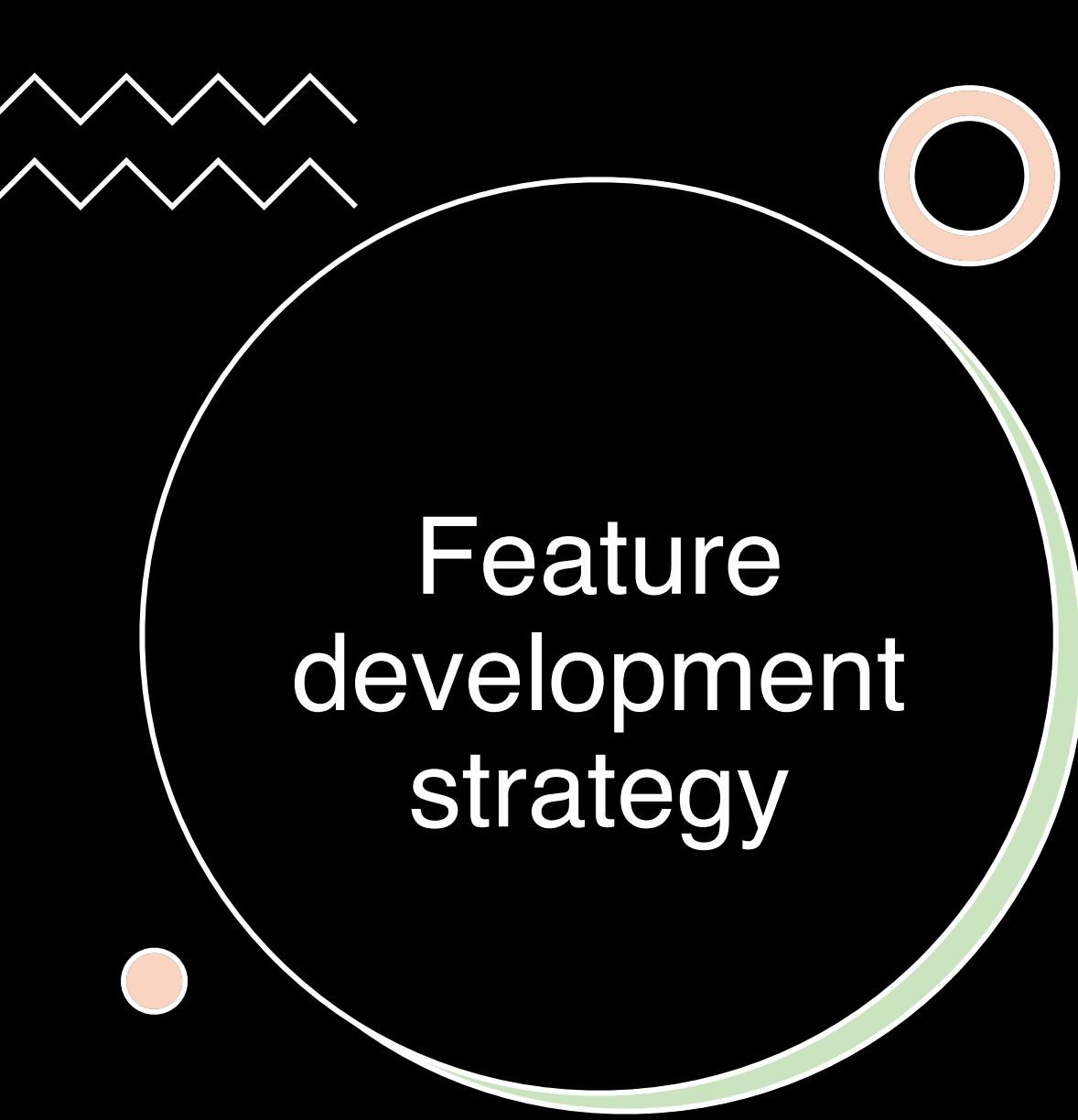
Architecture



CI/CD

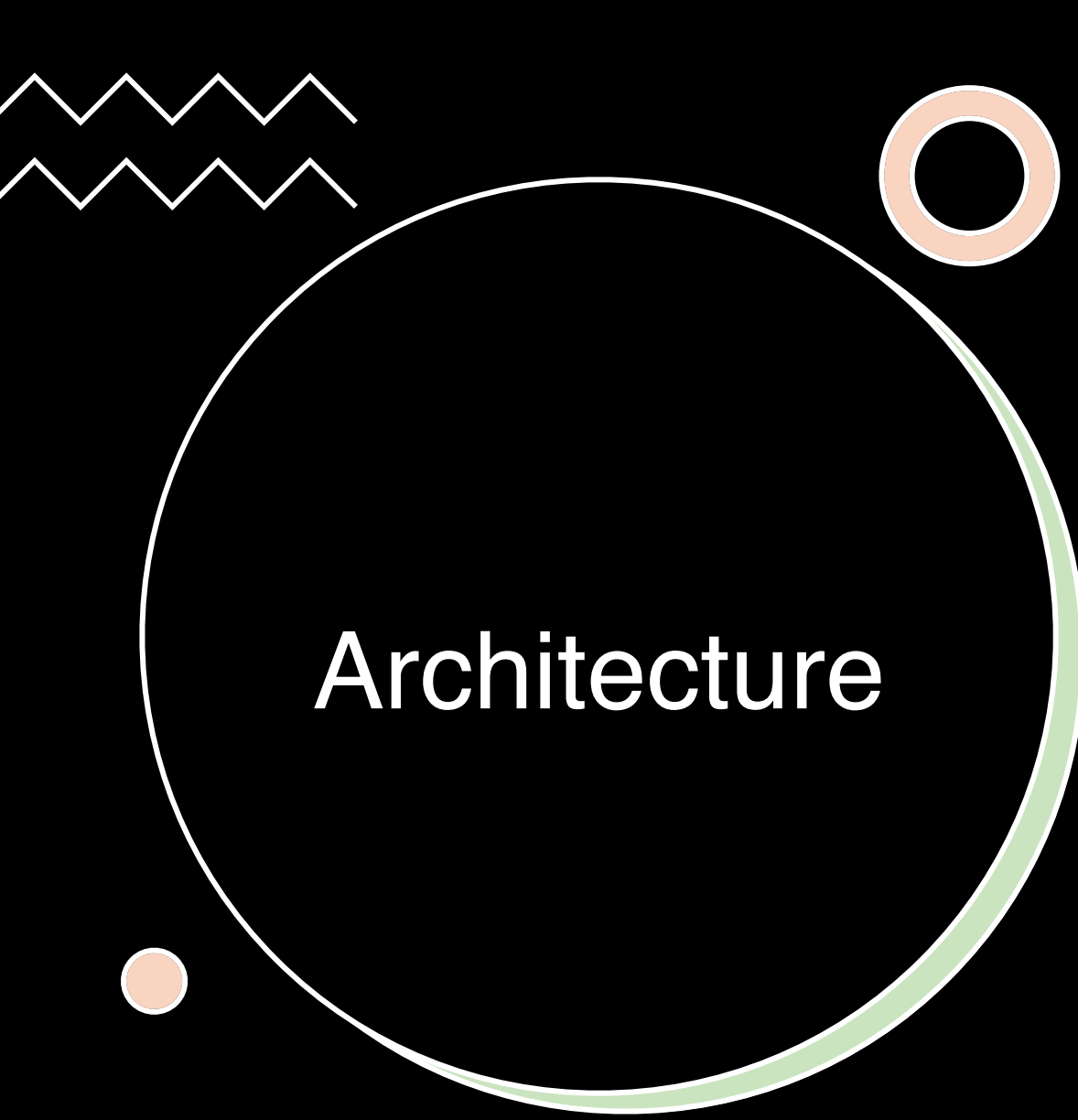


Set up database



Feature development strategy

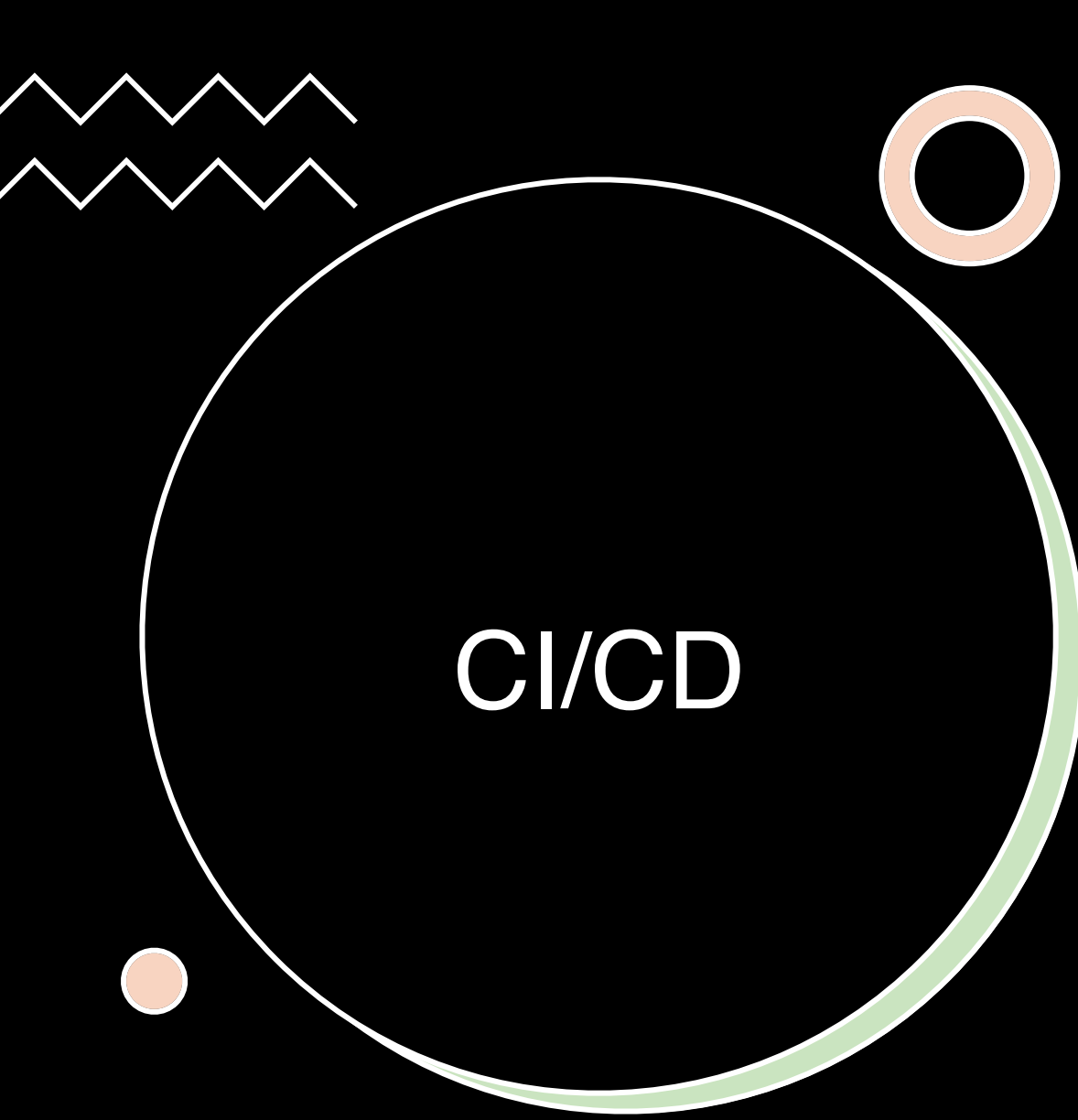
- Every person in the group focused on creating one core feature end to end
 - Helped enhance our skills across a variety front and back-end technologies
 - Balanced workload across the team
 - Git strategy: Feature branch
- 



Architecture

- NestJS for backend: fast development, modularization
- React for frontend: common framework
- Backend serve API and FE code
- Use MySQL for database
- Mono repository: API Dtos





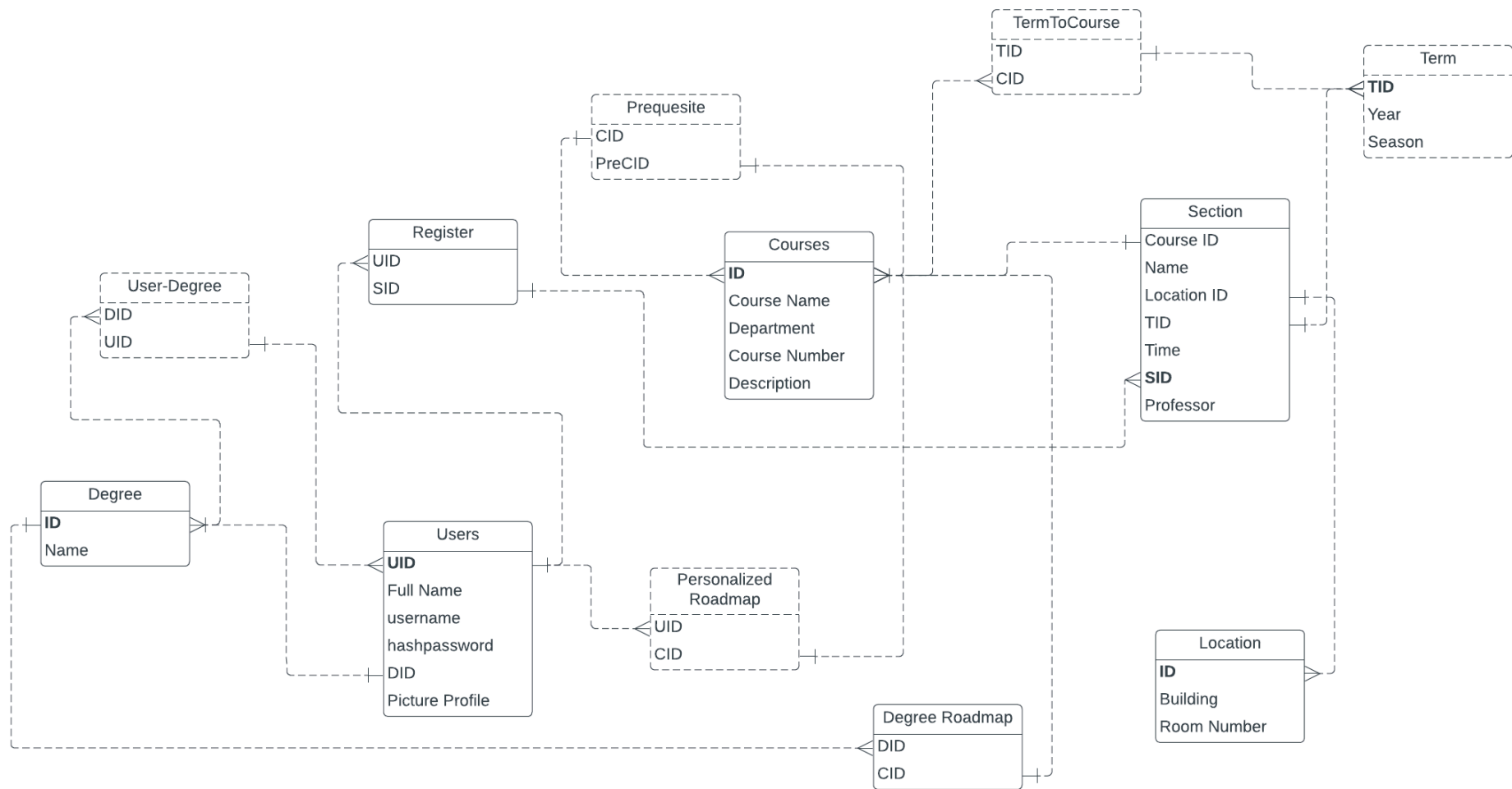
CI/CD

- Running on GitHub actions
 - Maintain the script as code
 - Docker for local development and CI/CD
 - Have tests suite running parallel
 - Having lint jobs to check code quality
- 



Set up database

- Design the relational schema
 - Convert schema to TypeORM entity
 - Provided seeding data to initialize database with data for local development
 - Running local database with Docker and production on Azure
- 



```
export class User {
  @PrimaryGeneratedColumn()
  uid: number;

  @Column()
  fullName: string;

  @Column( options: {unique: true})
  username: string;

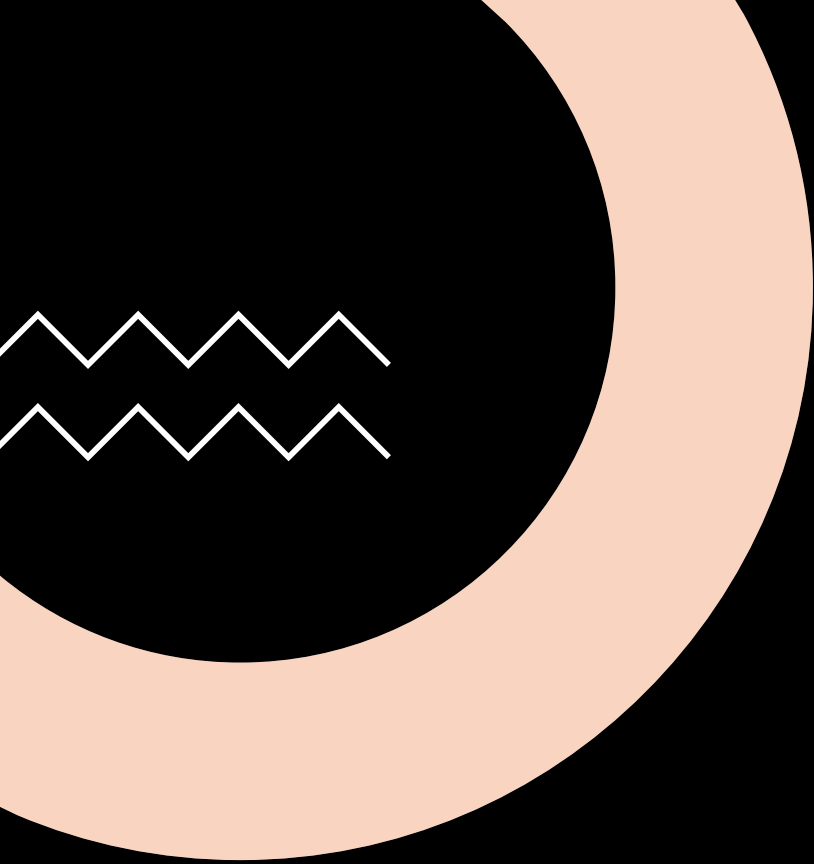
  @Column()
  hashPassword: string;

  @ManyToOne( typeFunctionOrTarget: () => Degree, inverseSide: (degree : Degree ) => degree.users)
  degree: Relation<Degree>;

  @ManyToMany( typeFunctionOrTarget: () => Section, inverseSide: (section : Section ) => section.users)
  sections: Relation<Section[]>;

  @ManyToMany( typeFunctionOrTarget: () => Course)
  @JoinTable( options: {name: 'planned_courses_user'})
  plannedCourses: Relation<Course[]>;

  @ManyToMany( typeFunctionOrTarget: () => Section, inverseSide: (section : Section ) => section.doneUsers)
  doneSections: Relation<Section[]>;
}
```



Reflections

- This project was a great chance for every one of us to collaborate in a team in an Agile methodology. It gives us a simulation on how to work in the industry
- This provides us chances to learn more about different technologies in computer science: front-end development, back-end development, devops





The end and Q/A

Thank you for listening